

Project Progress & Work Plan Schedule

Work Completed (Phases 0 – 5 | Up to Sep 1, 2026)

Week	Description of Work Done / Planned
Week 1 Jul 20 – Jul 26, 2026	• Finalized project scope, architecture blueprints, and technology stack (Spring Boot, PostgreSQL, React, Python FastAPI, Docker). • Set up repository structure: frontend/, backend/, ai-service/, and infra/ with Docker compose configuration. • Initialized Spring Boot backend skeleton (controller, service, repository, entity, dto packages) & PostgreSQL JPA connectivity. • Initialized React frontend (Vite + Tailwind CSS) and FastAPI microservice skeleton with placeholder health-check endpoint. • Configured GitHub Actions CI build pipeline and global exception handling with standard API response envelopes.
Week 2 Jul 27 – Aug 2, 2026	• Designed and created core PostgreSQL user database schema: users, roles, departments, students, faculty, and admins. • Implemented JWT-based authentication pipeline: Admin-provisioned signup, login, token refresh, and logout endpoints. • Configured BCrypt password hashing and secure password reset workflow. • Built Spring Security RBAC interceptors & annotations (@PreAuthorize) enforcing route and method-level access control. • Built Admin capability to create/deactivate student & faculty accounts, assign departments, and track audit logs. • Implemented role-gated frontend routing and baseline dashboards for Student, Faculty, and Admin portals.
Week 3 Aug 3 – Aug 9, 2026	• Designed and implemented Core Academic Data Model entities: Department, Course/Subject, Semester, Section, Enrollment, Timetable, Attendance, MarksInternal, MarksSemester. • Developed Admin CRUD REST APIs for structural academic entities (departments, courses, semesters, section slots). • Built Faculty-course mapping logic (linking faculty to subject and section allocations). • Implemented Student-course enrollment linkages and validation rules. • Created automated database seeding script generating realistic multi-department mock datasets for testing.
Week 4 Aug 10 – Aug 16, 2026	• Developed Student Portal backend APIs: subject-wise breakdown & overall attendance percentage calculations. • Implemented internal and semester marks query endpoints for student performance tracking. • Built automatic CGPA/SGPA calculation logic engine in Spring Boot service layer. • Implemented weekly timetable retrieval, fee status display module, and placement updates notice feed. • Integrated academic document generation service (marksheets and transcript PDF export) and notification feed.
Week 5 Aug 17 – Aug 23, 2026	• Developed Faculty Portal backend APIs: bulk attendance recording per session, subject, and section. • Implemented internal (CIA) and semester marks submission & update endpoints. • Built assignment upload & course material management endpoints (syllabus, study resources). • Created previous-year question paper upload staging interface for faculty submissions. • Implemented course-wise student performance analytics APIs (aggregating class averages, pass %, attendance %).

Week	Description of Work Done / Planned
Week 6 Aug 24 – Sep 1, 2026	• Developed Admin Portal feature logic: CSV bulk user import, profile editing, and account deactivation. • Built institutional report aggregation endpoints: department-wise results, pass %, backlog analysis, and attendance summaries. • Implemented system audit log viewer API tracking all administrative mutations. • Created question paper approval workflow (approve/reject moderation queue for faculty uploads). • Finalized database backup trigger endpoint and completed Phase 5 admin control surface verification.

Work to be Done (Phases 6 – 11 | Post Sep 1, 2026)

Week	Description of Work to be Done / Planned
Week 7 Sep 2 – Sep 8, 2026	• Develop Phase 6 Previous Year Question Paper Repository: search & filter APIs by subject, semester, department, and year. • Implement file storage integration (S3 bucket structure organized by dept/sem/course) and download tracking metrics.
Week 8 Sep 9 – Sep 15, 2026	• Develop Phase 7 Placement Management Module: company registration, drive posting, and job descriptions. • Implement student resume upload, application submission engine, and auto-filtering eligibility rules engine (CGPA/backlog cutoffs).
Week 9 Sep 16 – Sep 22, 2026	• Execute Phase 8 AWS Cloud Integration: deploy backend, frontend, and ai-service containers to EC2. • Migrate local PostgreSQL to Amazon RDS, configure S3 buckets, IAM policies, CloudWatch monitoring, and Nginx SSL certificates.
Week 10 Sep 23 – Sep 29, 2026	• Build Phase 9 AI Feature Integration (Part 1): FastAPI RAG engine, AI College Assistant chatbot, and Smart AI Search. • Train scikit-learn ML models for Student Performance Prediction (CGPA forecasting) and Attendance Risk Detection.
Week 11 Sep 30 – Oct 6, 2026	• Build Phase 9 AI Feature Integration (Part 2): AI Resume Analyzer (ATS scoring), AI Placement Assistant, and AI Notice Summarizer.
Week 12 Oct 7 – Oct 13, 2026	• Execute Phase 10 UI/UX Design & Full Frontend Build: implement Tailwind CSS design system, responsive dashboards, and interactive charts.
Week 13 Oct 14 – Oct 20, 2026	• Execute Phase 11 Testing & Security Audit: JUnit integration test suite, Postman regression suite, RBAC security audit, and load testing.
Week 14 Oct 21 – Oct 27, 2026	• Final Phase 11 Production Hardening: data migration tools, Swagger API documentation, admin runbook, pilot UAT, and final handover.